

# Causal Inference in Economics

## Difference-in-Differences (Conceptual & Storytelling Introduction)

ECON 4370

# The Central Question in Economics

# The Central Question in Economics

Much of economics is about one question:

**What is the causal effect of X on Y?**

# The Central Question in Economics

Much of economics is about one question:

**What is the causal effect of X on Y?**

Examples:

# The Central Question in Economics

Much of economics is about one question:

**What is the causal effect of X on Y?**

Examples:

- ▶ What is the effect of education on wages?

# The Central Question in Economics

Much of economics is about one question:

**What is the causal effect of X on Y?**

Examples:

- ▶ What is the effect of education on wages?
- ▶ What is the effect of a minimum wage increase on employment?

# The Central Question in Economics

Much of economics is about one question:

**What is the causal effect of X on Y?**

Examples:

- ▶ What is the effect of education on wages?
- ▶ What is the effect of a minimum wage increase on employment?
- ▶ What is the effect of pollution exposure on health?

# The Central Question in Economics

Much of economics is about one question:

## **What is the causal effect of X on Y?**

Examples:

- ▶ What is the effect of education on wages?
- ▶ What is the effect of a minimum wage increase on employment?
- ▶ What is the effect of pollution exposure on health?
- ▶ What is the effect of a new policy on economic outcomes?

# Correlation vs Causation

Suppose we observe:

$$Education \uparrow \rightarrow Wages \uparrow$$

# Correlation vs Causation

Suppose we observe:

$$Education \uparrow \rightarrow Wages \uparrow$$

Does this mean education **causes** higher wages?

# Correlation vs Causation

Suppose we observe:

$$Education \uparrow \rightarrow Wages \uparrow$$

Does this mean education **causes** higher wages?

Maybe...

# Correlation vs Causation

Suppose we observe:

$$Education \uparrow \rightarrow Wages \uparrow$$

Does this mean education **causes** higher wages?

Maybe...

But it could also be that:

# Correlation vs Causation

Suppose we observe:

$$\textit{Education} \uparrow \rightarrow \textit{Wages} \uparrow$$

Does this mean education **causes** higher wages?

Maybe...

But it could also be that:

- ▶ Ability affects both education and wages

# Correlation vs Causation

Suppose we observe:

$$\textit{Education} \uparrow \rightarrow \textit{Wages} \uparrow$$

Does this mean education **causes** higher wages?

Maybe...

But it could also be that:

- ▶ Ability affects both education and wages
- ▶ Family background affects both

# Correlation vs Causation

Suppose we observe:

$$\textit{Education} \uparrow \rightarrow \textit{Wages} \uparrow$$

Does this mean education **causes** higher wages?

Maybe...

But it could also be that:

- ▶ Ability affects both education and wages
- ▶ Family background affects both
- ▶ Motivation affects both

# Correlation vs Causation

Suppose we observe:

$$\textit{Education} \uparrow \rightarrow \textit{Wages} \uparrow$$

Does this mean education **causes** higher wages?

Maybe...

But it could also be that:

- ▶ Ability affects both education and wages
- ▶ Family background affects both
- ▶ Motivation affects both

**Correlation is not causation**

# The Fundamental Problem of Causal Inference

For each person (or place), two outcomes exist:

# The Fundamental Problem of Causal Inference

For each person (or place), two outcomes exist:

$$Y(1) = \text{Outcome if treated} \quad Y(0) = \text{Outcome if not treated}$$

# The Fundamental Problem of Causal Inference

For each person (or place), two outcomes exist:

$$Y(1) = \text{Outcome if treated} \quad Y(0) = \text{Outcome if not treated}$$

The causal effect is:

$$Y(1) - Y(0)$$

# The Fundamental Problem of Causal Inference

For each person (or place), two outcomes exist:

$$Y(1) = \text{Outcome if treated} \quad Y(0) = \text{Outcome if not treated}$$

The causal effect is:

$$Y(1) - Y(0)$$

But there is a problem.

# The Fundamental Problem of Causal Inference

For each person (or place), two outcomes exist:

$$Y(1) = \text{Outcome if treated} \quad Y(0) = \text{Outcome if not treated}$$

The causal effect is:

$$Y(1) - Y(0)$$

But there is a problem.

**We only observe one of them.**

# The Fundamental Problem of Causal Inference

For each person (or place), two outcomes exist:

$$Y(1) = \text{Outcome if treated} \quad Y(0) = \text{Outcome if not treated}$$

The causal effect is:

$$Y(1) - Y(0)$$

But there is a problem.

**We only observe one of them.**

We never observe both states for the same unit.

# The Ideal Experiment

The best way to estimate causal effects is:

# The Ideal Experiment

The best way to estimate causal effects is:

## **Randomized Experiments**

# The Ideal Experiment

The best way to estimate causal effects is:

## **Randomized Experiments**

Example:

# The Ideal Experiment

The best way to estimate causal effects is:

## **Randomized Experiments**

Example:

- ▶ Randomly assign some individuals to treatment

# The Ideal Experiment

The best way to estimate causal effects is:

## Randomized Experiments

Example:

- ▶ Randomly assign some individuals to treatment
- ▶ Randomly assign others to control

# The Ideal Experiment

The best way to estimate causal effects is:

## Randomized Experiments

Example:

- ▶ Randomly assign some individuals to treatment
- ▶ Randomly assign others to control
- ▶ Compare outcomes

# The Ideal Experiment

The best way to estimate causal effects is:

## Randomized Experiments

Example:

- ▶ Randomly assign some individuals to treatment
- ▶ Randomly assign others to control
- ▶ Compare outcomes

Randomization helps ensure:

$$\textit{Treatment} \perp \textit{Other Factors}$$

# The Ideal Experiment

The best way to estimate causal effects is:

## Randomized Experiments

Example:

- ▶ Randomly assign some individuals to treatment
- ▶ Randomly assign others to control
- ▶ Compare outcomes

Randomization helps ensure:

$$\textit{Treatment} \perp \textit{Other Factors}$$

But experiments are often impossible (or unethical) in economics.

# Natural Experiments

Sometimes the world creates situations that look like experiments.

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

Examples:

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

Examples:

- ▶ Policy changes

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

Examples:

- ▶ Policy changes
- ▶ Geographic variation

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

Examples:

- ▶ Policy changes
- ▶ Geographic variation
- ▶ Historical accidents

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

Examples:

- ▶ Policy changes
- ▶ Geographic variation
- ▶ Historical accidents
- ▶ Institutional rules

# Natural Experiments

Sometimes the world creates situations that look like experiments.  
Economists call these:

## Natural Experiments

Examples:

- ▶ Policy changes
- ▶ Geographic variation
- ▶ Historical accidents
- ▶ Institutional rules

One of the most famous early examples comes from the 1800s.

# The John Snow Cholera Study (London, 1854)

London experienced several deadly cholera outbreaks.

# The John Snow Cholera Study (London, 1854)

London experienced several deadly cholera outbreaks.  
Doctors believed cholera was caused by:

**Bad air ("miasma")**

# The John Snow Cholera Study (London, 1854)

London experienced several deadly cholera outbreaks.  
Doctors believed cholera was caused by:

**Bad air ("miasma")**

But physician **John Snow** had a different hypothesis:

# The John Snow Cholera Study (London, 1854)

London experienced several deadly cholera outbreaks.  
Doctors believed cholera was caused by:

**Bad air ("miasma")**

But physician **John Snow** had a different hypothesis:

Cholera spreads through **contaminated water**

# The John Snow Cholera Study (London, 1854)

London experienced several deadly cholera outbreaks.  
Doctors believed cholera was caused by:

**Bad air ("miasma")**

But physician **John Snow** had a different hypothesis:

Cholera spreads through **contaminated water**

How could he test this idea?

## Two Water Companies

In parts of London, households were served by two different water companies:

## Two Water Companies

In parts of London, households were served by two different water companies:

# Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**

## Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**
- ▶ **Lambeth Company**

# Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**
- ▶ **Lambeth Company**

The key difference:

# Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**
- ▶ **Lambeth Company**

The key difference:

- ▶ One company drew water from **polluted sections of the Thames**

# Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**
- ▶ **Lambeth Company**

The key difference:

- ▶ One company drew water from **polluted sections of the Thames**
- ▶ The other moved its intake **upstream to cleaner water**

# Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**
- ▶ **Lambeth Company**

The key difference:

- ▶ One company drew water from **polluted sections of the Thames**
- ▶ The other moved its intake **upstream to cleaner water**

Importantly (for identification):

# Two Water Companies

In parts of London, households were served by two different water companies:

- ▶ **Southwark & Vauxhall Company**
- ▶ **Lambeth Company**

The key difference:

- ▶ One company drew water from **polluted sections of the Thames**
- ▶ The other moved its intake **upstream to cleaner water**

Importantly (for identification):

**Households typically did not choose which company served them**

# A Natural Experiment

This created something very powerful:

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

But otherwise living in similar areas.

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

But otherwise living in similar areas.

Snow compared:

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

But otherwise living in similar areas.

Snow compared:

- ▶ Cholera deaths among households served by polluted water

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

But otherwise living in similar areas.

Snow compared:

- ▶ Cholera deaths among households served by polluted water
- ▶ Cholera deaths among households served by clean water

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

But otherwise living in similar areas.

Snow compared:

- ▶ Cholera deaths among households served by polluted water
- ▶ Cholera deaths among households served by clean water

Result:

# A Natural Experiment

This created something very powerful:

Two groups exposed to different water quality

But otherwise living in similar areas.

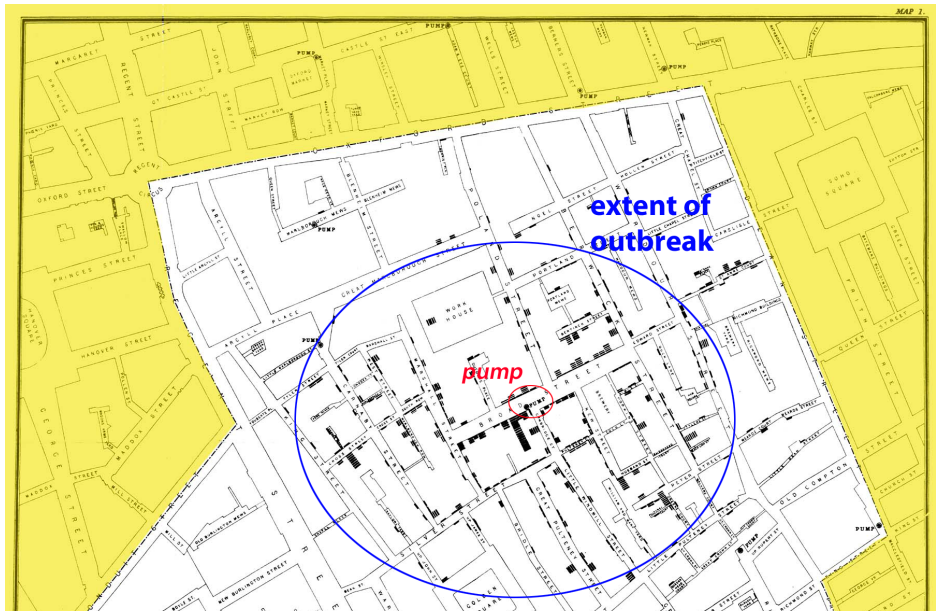
Snow compared:

- ▶ Cholera deaths among households served by polluted water
- ▶ Cholera deaths among households served by clean water

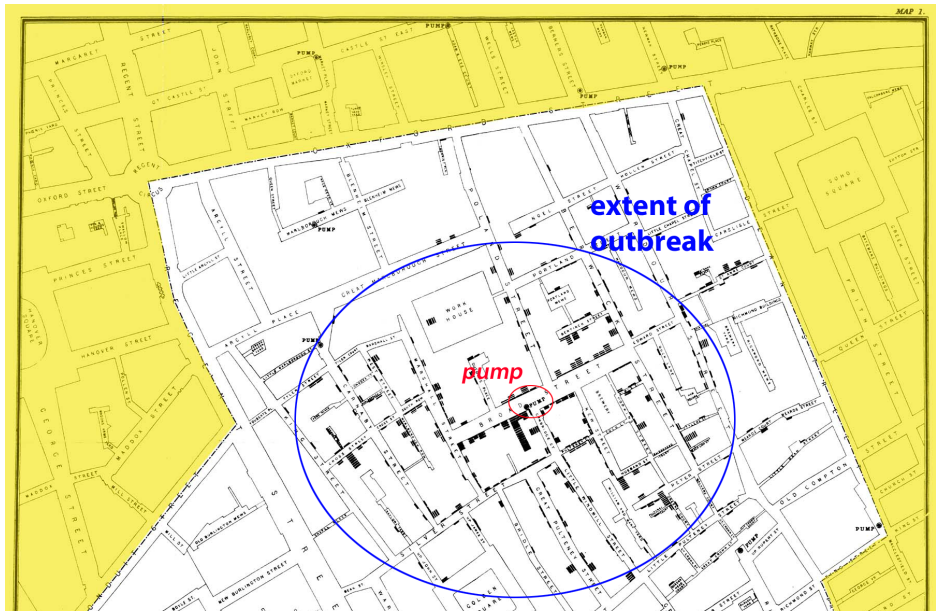
Result:

Cholera deaths were dramatically higher in polluted water areas.

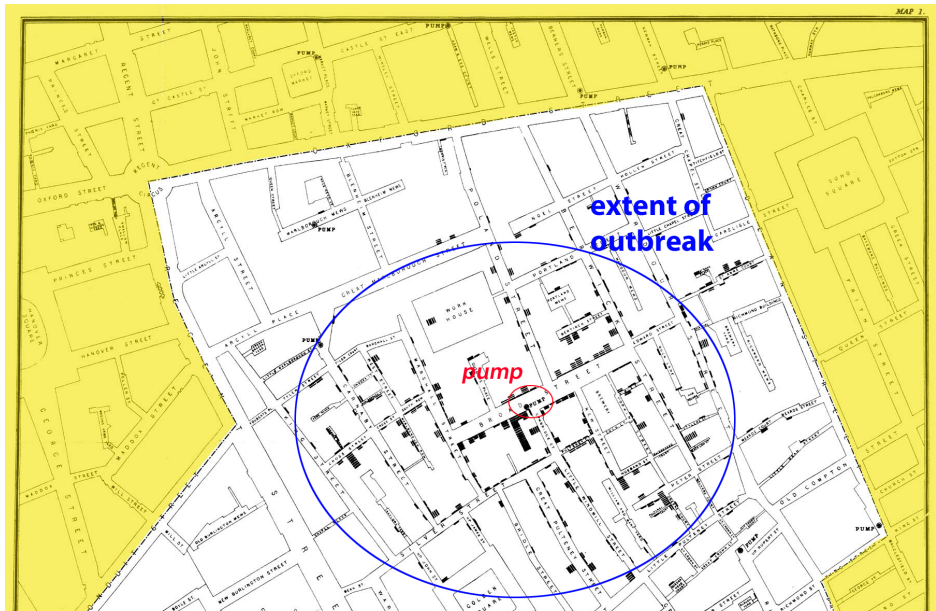
# John Snow's Cholera Map



# John Snow's Cholera Map



# John Snow's Cholera Map



# From John Snow to Modern Econometrics

John Snow compared:

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

Modern economists do something very similar:

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

Modern economists do something very similar:

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

Modern economists do something very similar:

- ▶ Compare a **treated group** to a **control group**

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

Modern economists do something very similar:

- ▶ Compare a **treated group** to a **control group**
- ▶ Observe outcomes **before and after** some change

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

Modern economists do something very similar:

- ▶ Compare a **treated group** to a **control group**
- ▶ Observe outcomes **before and after** some change
- ▶ Use the control group to estimate what would have happened otherwise

# From John Snow to Modern Econometrics

John Snow compared:

- ▶ Areas exposed to contaminated water
- ▶ Areas with cleaner water

Modern economists do something very similar:

- ▶ Compare a **treated group** to a **control group**
- ▶ Observe outcomes **before and after** some change
- ▶ Use the control group to estimate what would have happened otherwise

Difference-in-Differences formalizes this logic.

# Adding Time

Now suppose we observe:

# Adding Time

Now suppose we observe:

- ▶ A treated group

# Adding Time

Now suppose we observe:

- ▶ A treated group
- ▶ A control group

# Adding Time

Now suppose we observe:

- ▶ A treated group
- ▶ A control group
- ▶ Outcomes before the policy

# Adding Time

Now suppose we observe:

- ▶ A treated group
- ▶ A control group
- ▶ Outcomes before the policy
- ▶ Outcomes after the policy

# Adding Time

Now suppose we observe:

- ▶ A treated group
- ▶ A control group
- ▶ Outcomes before the policy
- ▶ Outcomes after the policy

This gives us a very useful structure.

# Adding Time

Now suppose we observe:

- ▶ A treated group
- ▶ A control group
- ▶ Outcomes before the policy
- ▶ Outcomes after the policy

This gives us a very useful structure.

## **Difference-in-Differences (DiD)**

## The Difference-in-Differences $2 \times 2$ Table

|                 | Before    | After     |
|-----------------|-----------|-----------|
| Control Group   | $Y_{C,0}$ | $Y_{C,1}$ |
| Treatment Group | $Y_{T,0}$ | $Y_{T,1}$ |

## The Difference-in-Differences 2×2 Table

|                 | Before    | After     |
|-----------------|-----------|-----------|
| Control Group   | $Y_{C,0}$ | $Y_{C,1}$ |
| Treatment Group | $Y_{T,0}$ | $Y_{T,1}$ |

First difference (time change):

$$\Delta_T = Y_{T,1} - Y_{T,0}$$

## The Difference-in-Differences 2×2 Table

|                 | Before    | After     |
|-----------------|-----------|-----------|
| Control Group   | $Y_{C,0}$ | $Y_{C,1}$ |
| Treatment Group | $Y_{T,0}$ | $Y_{T,1}$ |

First difference (time change):

$$\Delta_T = Y_{T,1} - Y_{T,0}$$

Second difference (time change in control):

$$\Delta_C = Y_{C,1} - Y_{C,0}$$

## The Difference-in-Differences 2×2 Table

|                 | Before    | After     |
|-----------------|-----------|-----------|
| Control Group   | $Y_{C,0}$ | $Y_{C,1}$ |
| Treatment Group | $Y_{T,0}$ | $Y_{T,1}$ |

First difference (time change):

$$\Delta_T = Y_{T,1} - Y_{T,0}$$

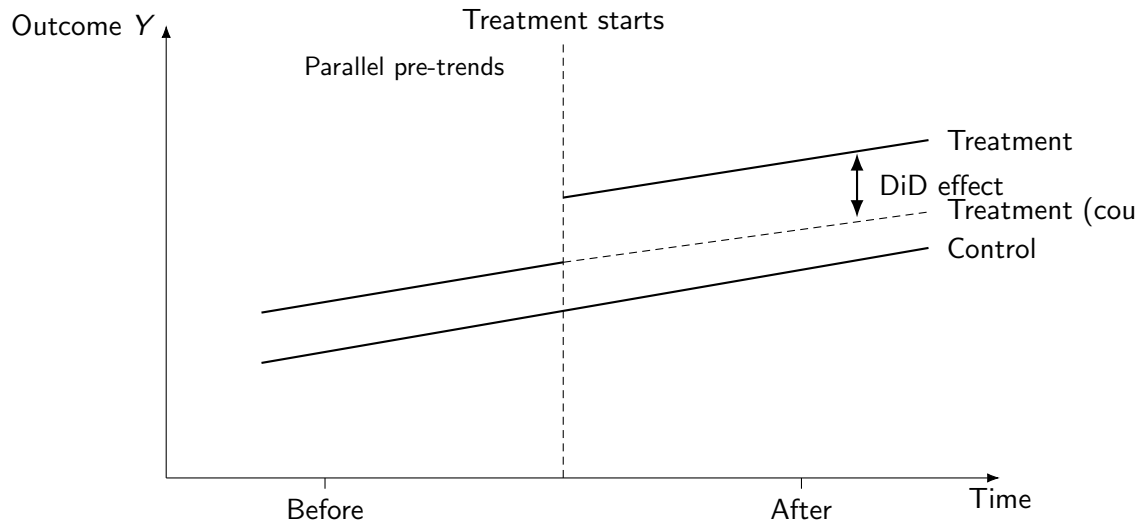
Second difference (time change in control):

$$\Delta_C = Y_{C,1} - Y_{C,0}$$

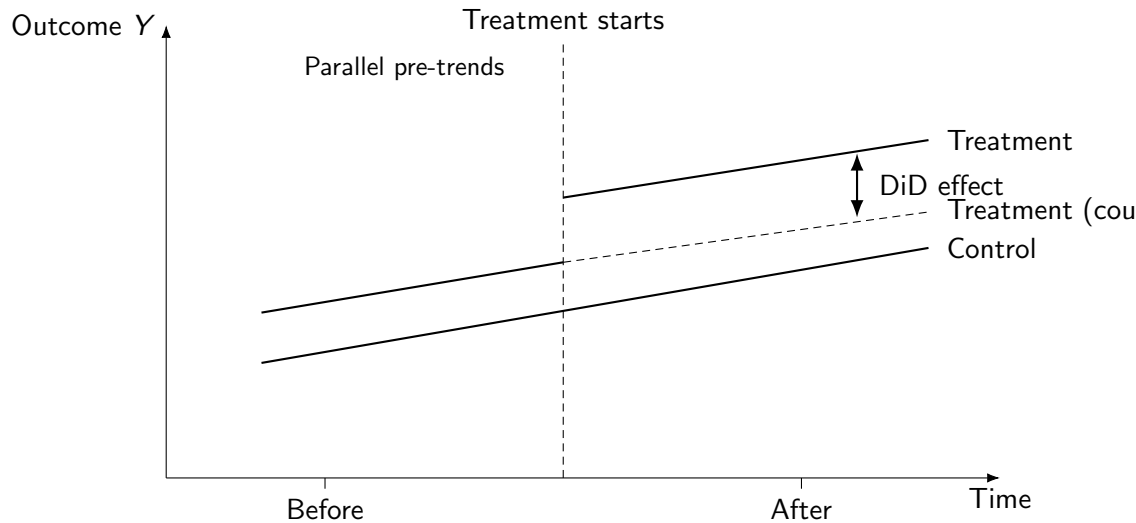
Difference-in-differences:

$$DiD = (Y_{T,1} - Y_{T,0}) - (Y_{C,1} - Y_{C,0})$$

## DiD Graphically (Clean Economics-Style Figure)



## DiD Graphically (Clean Economics-Style Figure)



The basic idea is to use the control trend to build the counterfactual for the treatment group.

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

**Parallel Trends**

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

## **Parallel Trends**

Without the treatment:

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

## **Parallel Trends**

Without the treatment:

Both groups would have evolved the same way over time.

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

## **Parallel Trends**

Without the treatment:

Both groups would have evolved the same way over time.

In words:

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

## **Parallel Trends**

Without the treatment:

Both groups would have evolved the same way over time.

In words:

- ▶ Groups can start at different levels

# The Key Assumption: Parallel Trends

Difference-in-Differences relies on one crucial assumption:

## Parallel Trends

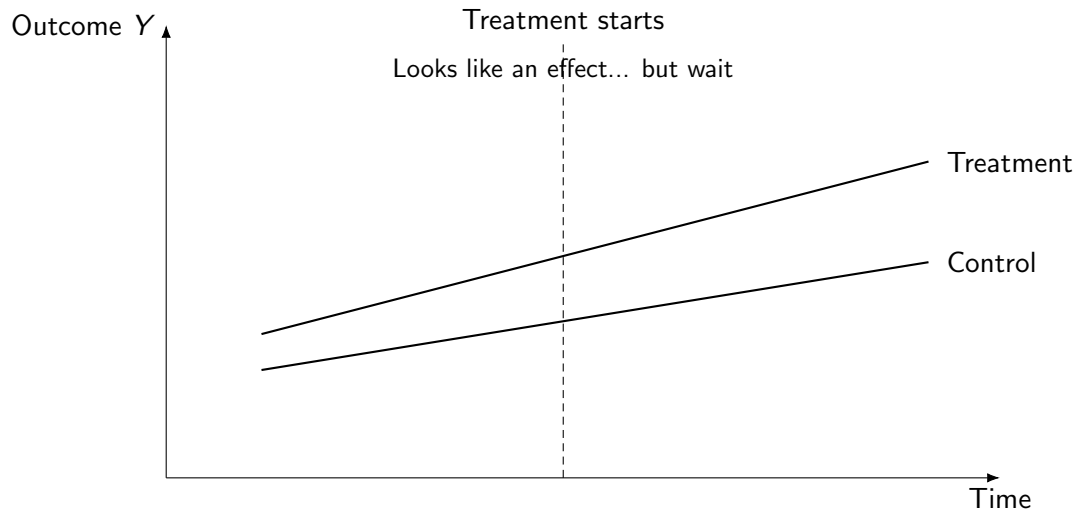
Without the treatment:

Both groups would have evolved the same way over time.

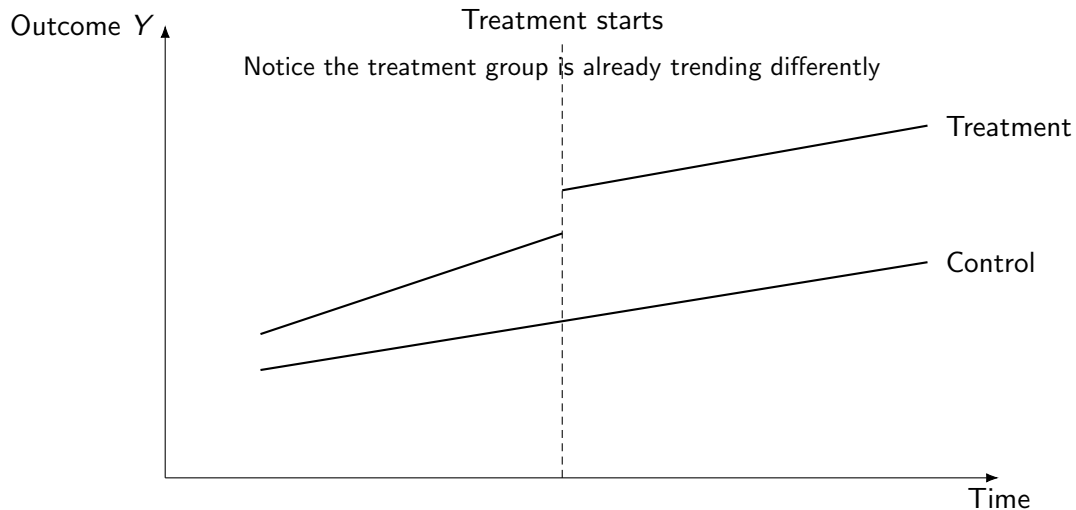
In words:

- ▶ Groups can start at different levels
- ▶ But their **trends** must be similar before treatment

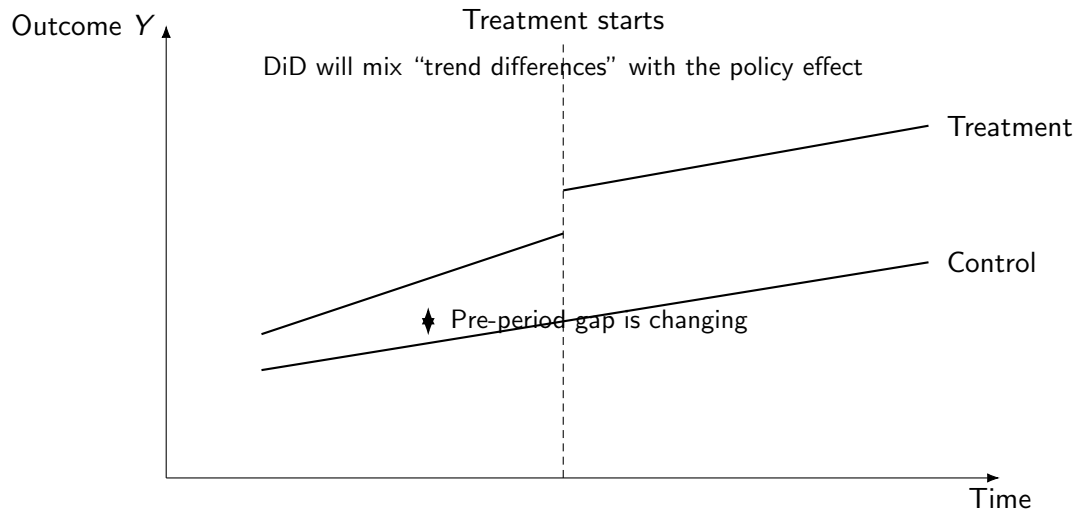
## When DiD Fails: Bad Parallel Trends (Animated)



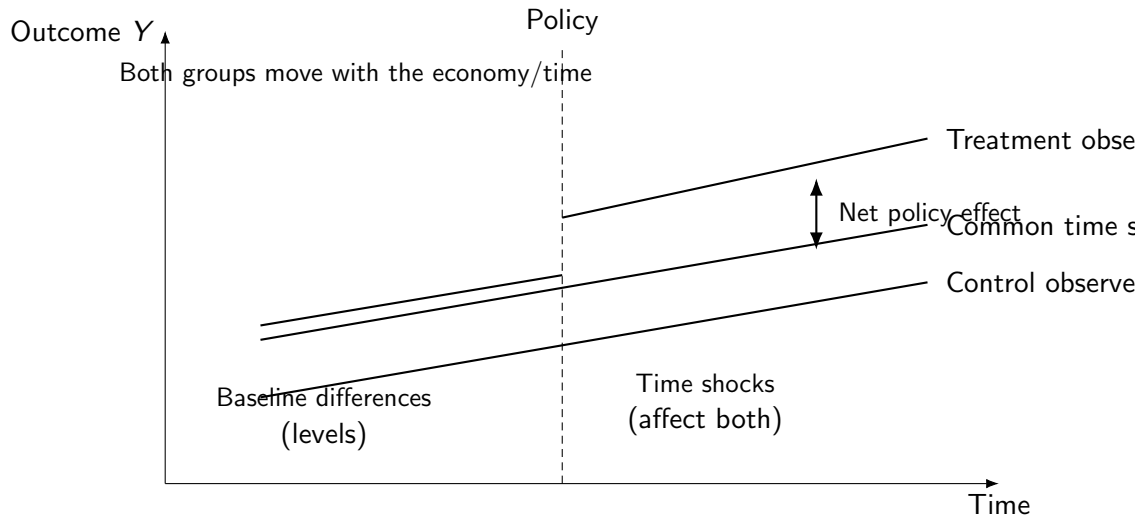
# When DiD Fails: Bad Parallel Trends (Animated)



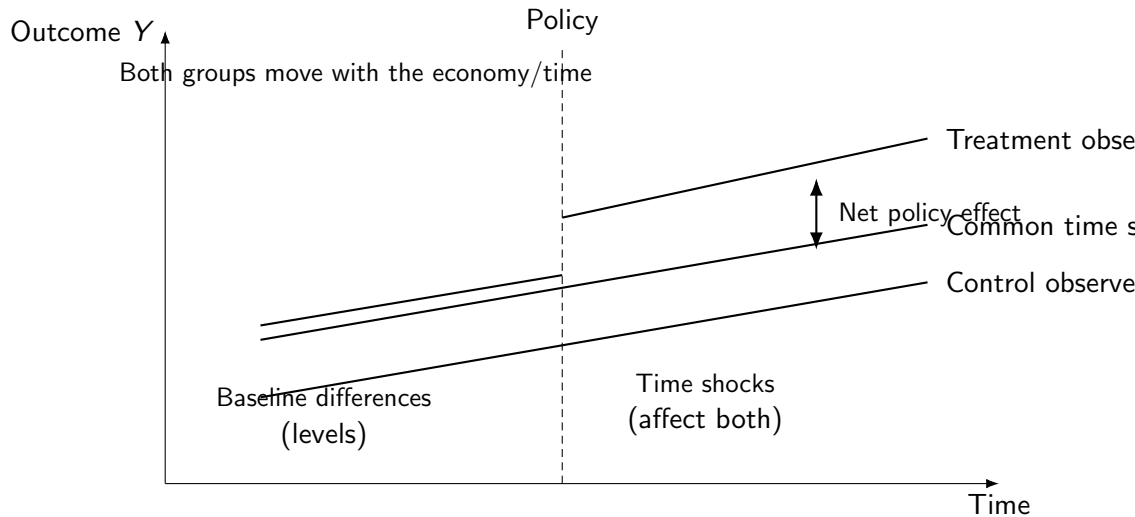
# When DiD Fails: Bad Parallel Trends (Animated)



# What DiD Removes (Why We Subtract Twice)

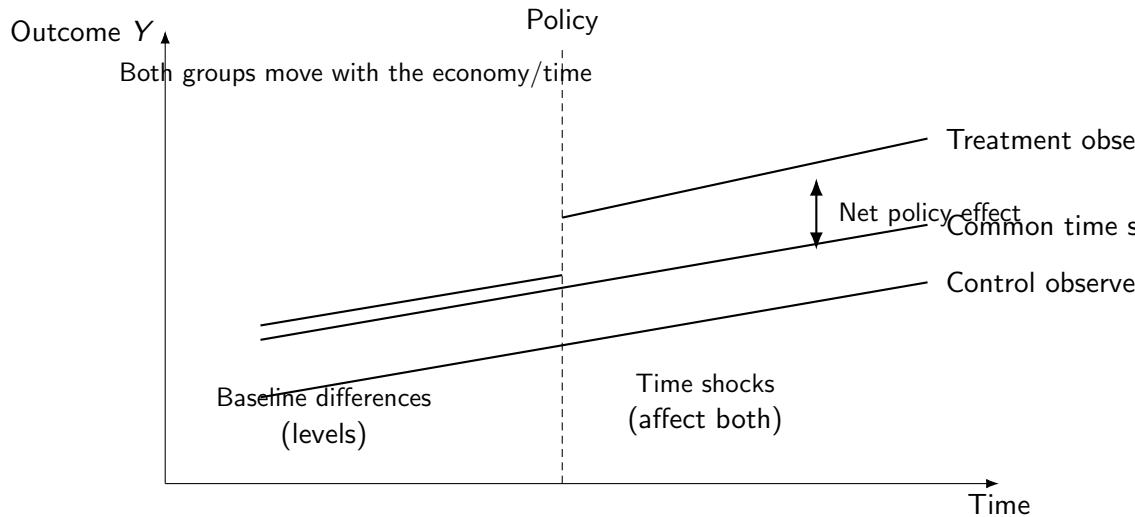


# What DiD Removes (Why We Subtract Twice)



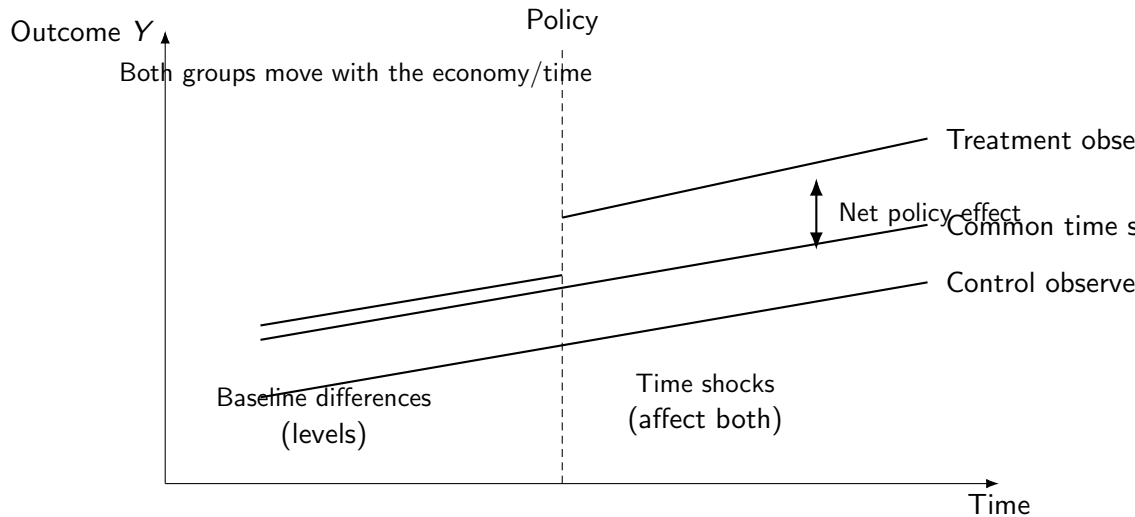
- First difference removes **baseline level differences** (treated vs control)

# What DiD Removes (Why We Subtract Twice)



- First difference removes **baseline level differences** (treated vs control)

# What DiD Removes (Why We Subtract Twice)



- First difference removes **baseline level differences** (treated vs control)

# Modern Example: Minimum Wage

A famous DiD application:

**Card and Krueger (1994)**

## Modern Example: Minimum Wage

A famous DiD application:

**Card and Krueger (1994)**

Question:

What is the effect of raising the minimum wage on employment?

## Modern Example: Minimum Wage

A famous DiD application:

**Card and Krueger (1994)**

Question:

What is the effect of raising the minimum wage on employment?

**Policy change:**

## Modern Example: Minimum Wage

A famous DiD application:

**Card and Krueger (1994)**

Question:

What is the effect of raising the minimum wage on employment?

**Policy change:**

- ▶ New Jersey increased its minimum wage (Treatment)

## Modern Example: Minimum Wage

A famous DiD application:

### **Card and Krueger (1994)**

Question:

What is the effect of raising the minimum wage on employment?

### **Policy change:**

- ▶ New Jersey increased its minimum wage (Treatment)
- ▶ Pennsylvania did not (Control)

## Modern Example: Minimum Wage

A famous DiD application:

### Card and Krueger (1994)

Question:

What is the effect of raising the minimum wage on employment?

#### Policy change:

- ▶ New Jersey increased its minimum wage (Treatment)
- ▶ Pennsylvania did not (Control)

Researchers compared:

|                        | Before     | After      |
|------------------------|------------|------------|
| Pennsylvania (Control) | Employment | Employment |
| New Jersey (Treatment) | Employment | Employment |

## Modern Example: Minimum Wage

A famous DiD application:

### Card and Krueger (1994)

Question:

What is the effect of raising the minimum wage on employment?

#### Policy change:

- ▶ New Jersey increased its minimum wage (Treatment)
- ▶ Pennsylvania did not (Control)

Researchers compared:

|                        | Before     | After      |
|------------------------|------------|------------|
| Pennsylvania (Control) | Employment | Employment |
| New Jersey (Treatment) | Employment | Employment |

DiD uses Pennsylvania to estimate the counterfactual for New Jersey.

# Summary

Today we built the intuition for DiD:

# Summary

Today we built the intuition for DiD:

# Summary

Today we built the intuition for DiD:

- ▶ Economics is about estimating **causal effects**

# Summary

Today we built the intuition for DiD:

- ▶ Economics is about estimating **causal effects**
- ▶ The fundamental problem: we never observe both  $Y(1)$  and  $Y(0)$

# Summary

Today we built the intuition for DiD:

- ▶ Economics is about estimating **causal effects**
- ▶ The fundamental problem: we never observe both  $Y(1)$  and  $Y(0)$
- ▶ Natural experiments can help (John Snow)

# Summary

Today we built the intuition for DiD:

- ▶ Economics is about estimating **causal effects**
- ▶ The fundamental problem: we never observe both  $Y(1)$  and  $Y(0)$
- ▶ Natural experiments can help (John Snow)
- ▶ DiD compares treatment vs control, before vs after

# Summary

Today we built the intuition for DiD:

- ▶ Economics is about estimating **causal effects**
- ▶ The fundamental problem: we never observe both  $Y(1)$  and  $Y(0)$
- ▶ Natural experiments can help (John Snow)
- ▶ DiD compares treatment vs control, before vs after
- ▶ The key assumption is **parallel trends**

# Summary

Today we built the intuition for DiD:

- ▶ Economics is about estimating **causal effects**
- ▶ The fundamental problem: we never observe both  $Y(1)$  and  $Y(0)$
- ▶ Natural experiments can help (John Snow)
- ▶ DiD compares treatment vs control, before vs after
- ▶ The key assumption is **parallel trends**

Next time: how to implement DiD using regression + real data (and how to test/defend parallel trends).